

Fernando Delgado-Chaves, PhD



AI Engineer & Computational Biologist

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 Hamburg, Germany EU citizen open to relocation available October 2026

PROFILE

I have strong analytical and problem-solving skills, with the ability to translate complex data into actionable business insights. I am continuously up-to-date with digital trends, emerging technologies, and the evolving AI landscape. Nine years as a computational biologist — network medicine, multi-omics, drug repurposing — then four years building **production agentic AI** on real scientific infrastructure. Most recently co-founder and CTO of an agentic platform turning scientific questions into validated, reproducible bioinformatics analyses, deployed into academic research groups and Biotech/Pharma R&D departments where data cannot leave the institution.

Co-Founder & CTO

Apr 2025 – Sep 2026

Galagos GmbH — Hamburg, Germany

- **Conceived Galagos during university research and co-founded the company**, translating the original concept into a production platform delivering executable scientific analyses, figures and reports. Defined the product vision and architecture through **Agile development**, using researcher interviews and usage analytics to prioritize reliability, onboarding and cost management. Engineered two generations of agent orchestration, automated scientific-skill curation and quality checks, developer tooling and end-to-end tests; co-developed the backend and interface, supported by reusable technical documentation and an internal knowledge base.
- **Established reproducible evaluation of scientific task performance** by building the BixBench execution and grading system, including numerical verification and resumable runs. Demonstrated **81.0% accuracy (166/205)** on the full public set and **92.0% on Verified-50**, at **~€0.87 per question** on the full-set run; published the evaluation results openly.
- **Initiated the company's sales efforts and led customer sales meetings**, combining technical discovery, tailored demonstrations and hands-on support with customer-specific ROI estimates and commercial proposals. Built **AI-agent sales workflows** for prospect discovery, qualification, personalized outreach, contact enrichment and CRM updates, automating reply detection, follow-up reminders and preparation of account briefs and proposal drafts. Supported adoption in academic and biotechnology research on EU-hosted infrastructure, reaching **450+ registered users and paying biotechnology customers**. Secured **~€234K in non-dilutive funding**.
- **Executed digital marketing workflows**, building LinkedIn content, branded visuals, product demonstrations and blog articles. Automated content preparation, media handling and search/social-preview metadata; developed analytics to refine messaging and content strategy. Grew **1,400+ net-new LinkedIn company-page followers and 40,284 organic impressions** over the year ending June 2026.

Junior Group Leader & Postdoctoral Researcher

May 2022 – Sep 2025

University of Hamburg, Institute of Computational Systems Biomedicine (CoSy.Bio) — Hamburg, Germany

- Appointed at age 28 to lead a group on **generative AI for biomedical research**. Directed interdisciplinary projects and secured **~€252K competitive funding** (Horizon Europe FLAME consortium, 2026; work package hosted at UHH). Presented work at international conferences, hackathons and workshops. **Taught** a 12-session graduate course, Introduction to / Applied Computational Systems Medicine.
- **Automated biomedical evidence processing** by engineering LLM agents for literature screening, extraction of disease, gene and treatment relationships, and execution of research workflows from natural-language questions. Converted unstructured data into ontology-mapped records for rare-disease research and **benchmarked 18 LLMs against human expert systematic reviews**.
- **Designed REPO4EU strategies to prioritize host-directed antiviral candidates** by integrating host-virus interactions, tissue expression and pathway evidence into candidate-gene selection, disease-module expansion and network-based drug ranking. Developed evidence-weighted seed scoring and viral-lifecycle-specific analyses; **evaluated 430 network modules** for HSV-1/2 research and assessed drug rankings against reference sets of approved and experimental treatments.
- **Developed and evaluated statistical methods to resolve molecular disease heterogeneity**, combining omics-based biclustering for patient stratification with differential co-expression networks to detect disrupted gene relationships beyond changes in individual gene expression. Contributed to improved recovery of breast-cancer and asthma subtypes; designed **DRaCOoN**, a differential-network method that **outperformed eight alternatives in benchmarking** and identified regulatory changes in three real-world transcriptomics datasets.

Predocotrual Researcher (PhD) & Research Assistant

Sep 2017 – May 2022

Pablo de Olavide University — Seville, Spain

- **Developed gene-network reconstruction algorithms** and implemented an ensemble-and-greedy approach in R/Java to identify gene co-expression relationships. **Used those to investigate molecular responses to DNA repair and SARS-CoV-2 infection** through network analysis; streamlined NGS workflows and mentored thesis students.
- **Grew a wet-lab foundation during three research stays**, applying CRISPR-Cas9, confocal microscopy and immunostaining to cancer models, genomic instability and developmental biology.

SELECTED SCIENTIFIC SOFTWARE & RESEARCH

ChatDRex — multi-agentic drug repurposing over biomedical knowledge graphs (Neo4j, integrating 20+ databases) and scientific literature; enhancement of **DrugRepoChatter**. Developed the original in LangGraph, KG / text embedding in FAISS, connected to third-party literature DB and computing services via API. Supervised the project; co-senior author.



AcadeMate — PRISMA-methodology agent for systematic literature reviews, released as a live service. Built the original Python pipeline with LangChain and custom solutions. Benchmarked performance against curated datasets.



DRaCOon — pathway-level differential regulation and co-expression analysis. Designed the algorithm using information theory and correlation-based methods with permutation tests. Numba-accelerated computing. Developed PyPI Python package. BMC Bioinformatics paper, PyPI release.



UnPaSt — unsupervised patient stratification by biclustering of omics data. Contributed to benchmarking against 22 methods on breast-cancer and asthma subtypes using every clustering algorithm in sci-kit learn.



nf-core/diseasemodulediscovery — Nextflow pipeline for network-based disease-module identification (with the REPO4EU consortium). Community-standard nf-core workflow benchmarking six module-detection tools.



LangChain — merged fix to streaming metadata in ChatAnthropicVertex (#1263). Contribution accepted into a major AI framework.



EDUCATION

- **PhD, Computational Biology** — Pablo de Olavide University, Seville (2020–2023). **Summa cum laude**. Thesis: new algorithms for gene network reconstruction.
- **MSc, Advanced Bioinformatic Analysis** (9.14/10), **MSc, Health Biotechnology**, Cell Therapy (9.56/10) and **MSc, Higher Education** (teaching qualification, content and language integrated learning (CLIL); 8.81/10) — Pablo de Olavide University (2018–2021).
- **BSc, Biotechnology (bilingual)** — Pablo de Olavide University (2014–2018). 8.82/10. Erasmus year, Ruhr University Bochum, Germany.
- **Supplementary education** — **AI4MedChem Summer School**, University of Münster (2024): small molecules, cheminformatics, GNNs/transformers for drug design • **GTIPI Summer School**, IMBEI Mainz (2022): genomics, transcriptomics, proteomics and medical informatics • **EUGLOH Workshop**, UKE Hamburg (2025): machine learning for biomedical applications • **AI.STARTUP.HUB Ideation programme**, Hamburg Innovation (2025): positioning and narrative, digital sales, startup financing and pricing, investor readiness and pitch, value-driven scaling • **DeepLearning.AI** agent courses (2024): LangGraph, AutoGen, crewAI • **IBM Data Science Professional Certificates** (2020).

SELECTED PUBLICATIONS

- **Delgado-Chaves, F. M.**, et al. Transforming literature screening: the emerging role of large language models in systematic reviews. **PNAS** 122(2):e2411962122 (2025).
- **Delgado-Chaves, F. M.**, et al. DRaCOon: pathway-level differential co-expression analysis in transcriptomics. **BMC Bioinformatics** 26(1):137 (2025).
- Kuehl, M., ... **Delgado-Chaves, F. M.**, et al. Pathology-oriented multiplexing enables integrative disease mapping (PathoPlex). **Nature** 644:516–526 (2025).
- Süwer, S.*, ... **Delgado-Chaves, F. M.**†, Maier, A.† Conversational multi-agentic disease-module identification and drug repurposing with ChatDRex. Under review (2025). **Co-senior author.**
- Santamaría-Gómez, J., Rubio, M. Á., ... **Delgado-Chaves, F. M.**, et al. Role of a cryptic tRNA gene operon in survival under translational stress. **Nucleic Acids Research** 49(15):8757–8776 (2021).
- **Delgado, F. M.**, Gómez-Vela, F. Computational methods for gene regulatory networks reconstruction and analysis: a review. **Artificial Intelligence in Medicine** (2019).

SKILLS

Bioinformatics

Network medicine • Drug repurposing • Disease-module identification • Molecular biology • Statistics and survival analysis • Motif discovery • Clustering and enrichment analysis • Multi-omics • NGS • Nextflow • Knowledge graphs • Data-driven decision making

Agentic AI

Agentic (scientific) system design • RAG • LLM frameworks (LangChain, LlamalIndex, AutoGen, CrewAI) • Multi-agent orchestration • LLM fine-tuning (Unsloth) • AI agents observability (Langfuse/LangSmith) • LLM evaluation & benchmarking • Harness/Context/Prompt engineering • Vector Databases • Multi-modal embeddings • Coding Agents proficiency (Claude Code, Codex, Cursor)

Programming & Data

Python • R • Bash • SQL/NoSQL • (Un)Structured data integration • Data Science & Visualization • Machine Learning • Deep Learning • Graph/network analysis/GNN • Statistics • HPC • GPU (CUDA) • Reproducible analysis & audit trails • Privacy-preserving data analysis • Docker • CI/CD • GitHub workflows • Forward-deployed delivery

LANGUAGES

Spanish (Native) • **English** (C1) • **German** (B1) • **French** (B1) • **Portuguese** (B1) • **Italian** (B1)

RESEARCH

443 citations • **h-index 8** • **i10 5** (Google Scholar). Peer reviewer for **4 journals and 3 conferences**. Scientific supervision of students: **14 MSc, 4 BSc, 1 PhD** • 3 PhD-thesis committees.

RECOGNITION

Young Investigator Award & Best Talk, REPO4EU (2023) • Top 3 Speaker, REXPO'22. **Antler** residency Berlin 2025, **one of ~100 founders selected from ~10,000 applicants**. **Deep Tech Momentum 2026 (DTM100)** cohort • **Falling Walls Venture 2026** nominee • **NVIDIA Inception**. Creator of the **“Your Research, Supercharged”** AI-literacy seminar in scientific production (taught at 10+ institutions and research groups) • **Head of Communications**, Spanish Society of AI in Biomedicine (IABioMed, 2025).